

SQL Part 9: Database Design Made Easy (Normalization, 1NF 2NF 3NF, ER Diagrams) — Free Practice Guide

Beyond School · Ready to learn in a fun way! Anyone can learn — easy! Nothing to install, just a web browser.

Step 1: Open a free online SQL playground

- sqliteonline.com — choose "SQLite", paste, click Run.
- db-fiddle.com — pick "SQLite" as the database, paste and Run.
- sqlime.org — a simple SQLite editor in your browser.
- sqlbolt.com — free interactive lessons for extra practice.

Step 2: Paste this setup script and click Run

It creates the same library tables used in the video. If anything goes wrong, just run it again in a fresh playground.

```
CREATE TABLE books (
  id INTEGER PRIMARY KEY,
  title TEXT,
  author TEXT,
  year INTEGER,
  genre TEXT,
  pages INTEGER
);
INSERT INTO books VALUES
  (1, 'Harry Potter', 'J.K. Rowling', 1997, 'Fantasy', 309),
  (2, 'Matilda', 'Roald Dahl', 1988, 'Comedy', 240),
  (3, 'Wonder', 'R.J. Palacio', 2012, 'Drama', 310),
  (4, 'The Hobbit', 'J.R.R. Tolkien', 1937, 'Fantasy', 310),
  (5, 'Holes', 'Louis Sachar', 1998, 'Mystery', 233),
  (6, 'Charlotte''s Web', 'E.B. White', 1952, 'Classic', 192),
  (7, 'Dog Man', 'Dav Pilkey', 2016, 'Comedy', 240),
  (8, 'Coraline', 'Neil Gaiman', 2002, 'Fantasy', 162);

CREATE TABLE members (
  member_id INTEGER PRIMARY KEY,
  name TEXT,
```

```
  city TEXT,
  joined TEXT,
  email TEXT
);
INSERT INTO members VALUES
  (1, 'Aisha', 'Austin', '2023-01-15', 'aisha@mail.com'),
  (2, 'Ben', 'Denver', '2023-03-02', NULL),
  (3, 'Carlos', 'Austin', '2024-06-20', 'carlos@mail.com'),
  (4, 'Dana', 'Boston', '2024-09-10', NULL),
  (5, 'Eli', 'Denver', '2025-02-01', 'eli@mail.com');

CREATE TABLE loans (
  loan_id INTEGER PRIMARY KEY,
  book_id INTEGER,
  member_id INTEGER,
  loan_date TEXT,
  return_date TEXT
);
INSERT INTO loans VALUES
  (1, 1, 1, '2026-01-05', '2026-01-19'),
  (2, 3, 1, '2026-02-10', NULL),
  (3, 2, 2, '2026-01-20', '2026-02-03'),
  (4, 4, 3, '2026-03-01', '2026-03-15'),
  (5, 1, 3, '2026-04-02', NULL),
  (6, 7, 4, '2026-04-10', '2026-04-24'),
  (7, 7, 2, '2026-05-01', NULL),
  (8, 5, 4, '2026-05-15', NULL);
```

Step 3: Run every query from the video, in order

Paste one at a time, click Run, and compare with the expected result.

A messy table

```
CREATE TABLE messy (loan_id INTEGER,    member TEXT, city TEXT, books TEXT); INSERT INTO messy VALUES    (1, 'Aisha', 'Austin', 'Harry Potter, Wonder'),    (2, 'Ben', 'Denver', 'Matilda, Dog Man'),   (3, 'Carlos', 'Austin', 'The Hobbit, Harry Potter');
SELECT * FROM messy;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: messy
```

The problem with lists

```
SELECT * FROM messy WHERE books LIKE '%Potter%';
```

```
Error message (expected):
error
no such table: messy
```

Fixing it: 1NF

```
CREATE TABLE visit_books (    loan_id INTEGER, member TEXT, book TEXT); INSERT INTO visit_books VALUES   (1, 'Aisha', 'Harry Potter'), (1, 'Aisha', 'Wonder'),   (2, 'Ben', 'Matilda'), (2, 'Ben', 'Dog Man'),   (3, 'Carlos', 'The Hobbit'), (3, 'Carlos', 'Harry Potter');
SELECT * FROM visit_books;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: visit_books
```

Now counting works

```
SELECT book, COUNT(*) AS times FROM visit_books GROUP BY book ORDER BY times DESC, book;
```

```
Error message (expected):
error
no such table: visit_books
```

3NF: a cities table

```
CREATE TABLE cities (city TEXT PRIMARY KEY, state TEXT); INSERT INTO cities VALUES ('Austin', 'Texas'),   ('Denver', 'Colorado'), ('Boston', 'Massachusetts');
SELECT m.name, m.city, c.state FROM members AS m JOIN cities AS c ON m.city = c.city;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: cities
```

One-to-many in SQL

```
SELECT m.name, COUNT(l.loan_id) AS loans FROM members AS m LEFT JOIN loans AS l ON m.member_id = l.member_id GROUP BY m.name ORDER BY m.name;
```

```
Expected result:
name | loans
Aisha | 2
Ben | 2
Carlos | 2
Dana | 2
Eli | 0
```

Many-to-many: authors

```
CREATE TABLE authors (author_id INTEGER PRIMARY KEY, name TEXT); INSERT INTO authors VALUES (1, 'Neil Gaiman'),    (2, 'Terry Pratchett'), (3, 'Roald Dahl'); INSERT INTO books VALUES (9, 'Good Omens',   'Gaiman & Pratchett', 1990, 'Fantasy', 412);
SELECT * FROM authors;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: authors
```

The junction table

```
CREATE TABLE book_authors (   book_id INTEGER REFERENCES books(id),   author_id INTEGER REFERENCES authors(author_id),   PRIMARY KEY (book_id, author_id)); INSERT INTO book_authors VALUES   (8, 1), (9, 1), (9, 2), (2, 3);
SELECT * FROM book_authors;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: book_authors
```

Querying many-to-many

```
SELECT b.title, a.name FROM book_authors AS ba JOIN books AS b ON ba.book_id = b.id JOIN authors AS a ON ba.author_id = a.author_id ORDER BY b.title, a.name;
```

```
Error message (expected):
```

```
error
no such table: book_authors
```

Counting across the bridge

```
SELECT a.name, COUNT(*) AS books FROM authors AS a JOIN book_authors AS ba ON a.author_id = ba.author_id GROUP BY a.name ORDER BY books DESC, a.name;
```

```
Error message (expected):
error
no such table: authors
```

Step 4: Challenges (try first!)

C1. Practice the bridge: add Terry Pratchett as a pretend co-author of Coraline, list all pairs, then remove that row.
C2. Add 'Chicago, Illinois' to cities and list members with their state (keep members with no match).
C3. Which authors have written more than one book?

Answers

C1. Practice the bridge: add Terry Pratchett as a pretend co-author of Coraline, list all pairs, then remove that row.

```
INSERT INTO book_authors VALUES (8, 2);
SELECT b.title, a.name FROM book_authors AS ba JOIN books AS b ON ba.book_id = b.id JOIN authors AS a ON ba.author_id = a.author_id;
DELETE FROM book_authors WHERE book_id = 8 AND author_id = 2;
```

```
Error message (expected):
error
no such table: book_authors
```

```
Error message (expected):
error
no such table: book_authors
```

```
Error message (expected):
error
no such table: book_authors
```

C2. Add 'Chicago, Illinois' to cities and list members with their state (keep members with no match).

```
INSERT INTO cities VALUES ('Chicago', 'Illinois');
SELECT m.name, c.state FROM members AS m LEFT JOIN cities AS c ON m.city = c.city;
```

```
Error message (expected):
error
no such table: cities
```

```
Error message (expected):
error
no such table: cities
```

C3. Which authors have written more than one book?

```
SELECT a.name FROM authors AS a JOIN book_authors AS ba ON a.author_id = ba.author_id
GROUP BY a.name HAVING COUNT(*) > 1;
```

```
Error message (expected):
error
no such table: authors
```

Keep going!

Watch the whole free SQL course on Beyond School, starting with SQL Part 1:

https://youtu.be/72JvYu2OtiA
Beyond School — Ready to learn in a fun way!